

The called strike zone shrank twice, and drifted with no rule change

Introduction

The called strike zone shrank in 2026, when regular-season ABS challenges began. Umpire grading had already tightened in 2025, from a buffer two inches outside the edge to three-quarters of an inch either side. Published estimates of the 2026 change start from 2025 or fold it into a trend (Lee et al., arXiv, 22 September 2026), so none gives both steps intervals against old-rule seasons. This paper measures both steps against 2022-2024 and finds the 2026 step the larger, although a failed placebo leaves both descriptive.

Methods

To put five seasons on one zone, I re-projected each 2022-2025 pitch to the plate midpoint Statcast adopted in 2026. I fitted called-strike probability near the edge, from 1,830,231 called pitches in 12,060 regular-season games, and read each season's zone as its 50 percent contour. With ABS-measured heights, the zone geometry matched 10,164 of 10,168 challenge verdicts. I measured both steps net of the 2022-2024 trend, pooled umpires, and pre-registered a placebo between 2023 and 2024, two seasons under one rule.

Results

The placebo failed: with no rule change, the zone grew 10.8 (90% CI 7.7 to 13.9) square inches, beyond its ± 3 margin, so every step below is descriptive. As Figure 1 shows, the 2025 step was 23.4 (95% CI 18.6 to 28.3) square inches. The 2026 step, 33.1 (95% CI 28.7 to 37.5), exceeded every All-Star-break swing, the largest 6.8 square inches. As Table 1 shows with 95% intervals, the 2025 step came mostly from the sides, each in by 0.41 inches. In 2026 the bottom rose 0.65 inches and the top fell 0.63, although the top's 95% interval reads slightly too narrow. With 2025 at the plate front, as in the published 8 to 22 (Clemens, FanGraphs, 28 April 2026), the 2025-to-2026 loss is 20.0 (95% CI 13.0 to 28.2) square inches. With both seasons at mid-plate it is 32.8 (95% CI 29.3 to 36.6), so the plate-front reading hides part of it. Umpires moved together in 2026, with a between-umpire spread of 0.04 (95% CI 0.00 to 0.10) inches.

Conclusion

The zone had already shrunk in 2025, before regular-season challenges began, so framing and pitch-location comparisons that start from 2025 miss that step. The 2026 step is still the larger, at 59 percent of the two combined (95% CI 53 to 65). The contribution is both steps with intervals against three old-rule seasons, the between-umpire spread of the 2026 shift, and the 2023-to-2024 drift, a caveat for any two-season comparison. The limit is the failed placebo: the design cannot assign either step to its rule. A binned cross-check misses its pre-registered ± 3 square inch tolerance, finding both steps 6.0 and 7.6 square inches larger. Games from 22 September 2026 and the postseason are sealed for a later calibration check. The sealed-set definition and acceptance criteria were committed publicly before the sealed set was opened.

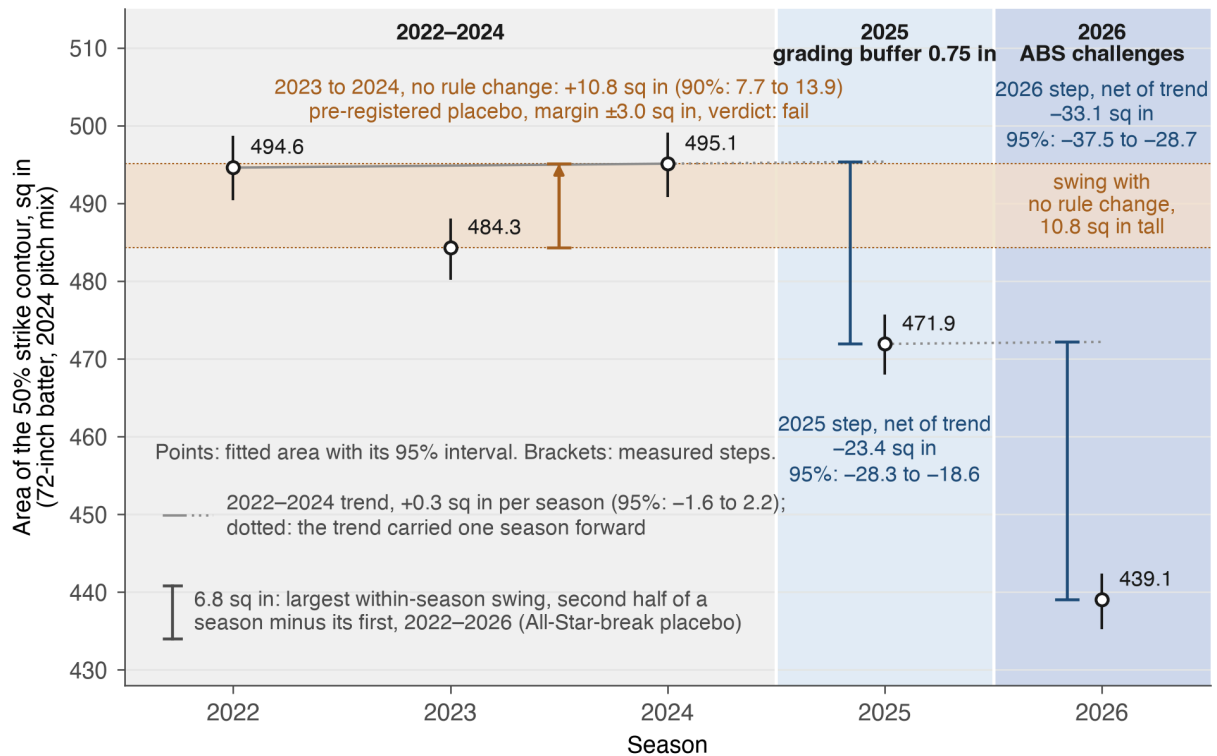


Figure 1. Area of the 50 percent strike contour, 2022–2026 (72-inch batter, 2024 pitch mix), 95% intervals. Steps net of the 2022–2024 trend: 2025, –23.4 sq in; 2026, –33.1 sq in. From 2023 to 2024, with no rule change, the area rose 10.8 sq in (90%: 7.7 to 13.9), failing the pre-registered placebo; the shaded band shows that swing.

Change in the fitted 50 percent contour	Top edge (in)	Bottom edge (in)	Half-width (in)	Area (sq in)	Shadow-band strike rate (pp)
2022-2024 trend, per season	0.13 [0.09, 0.18]	0.06 [0.02, 0.09]	-0.03 [-0.06, -0.01]	0.3 [-1.6, 2.2]	-0.09 [-0.41, 0.24]
2023 to 2024, no rule change (pre-registered placebo, 90% interval; margins ± 3 sq in and ± 0.5 pp): FAILED	—	—	—	10.8 [7.7, 13.9] 90% interval	1.54 [1.03, 2.04] 90% interval
2025 step, net of trend (grading buffer 0.75 in)	-0.18 [-0.32, -0.05]	0.18 [0.08, 0.29]	-0.41 [-0.49, -0.33]	-23.4 [-28.3, -18.6]	-4.46 [-5.28, -3.63]
2026 step, net of trend (ABS challenges)	-0.63 [-0.76, -0.50]	0.65 [0.55, 0.74]	-0.25 [-0.32, -0.16]	-33.1 [-37.5, -28.7]	-6.05 [-6.81, -5.31]
2024 to 2026, total	-0.54 [-0.65, -0.43]	0.94 [0.85, 1.03]	-0.72 [-0.80, -0.64]	-56.0 [-59.6, -52.0]	-10.69 [-11.28, -10.04]
Plate-plane component, 2025 read at mid-plate minus at the front	-0.77 [-0.94, -0.61]	-1.11 [-1.24, -0.98]	0.12 [-0.01, 0.25]	12.9 [6.3, 19.2]	—

Cells are point [95% interval] from coefficient draws of the fitted model; the placebo row prints its pre-registered 90% interval. A positive value moves an edge up or outward, or enlarges the area. Because a season pair with no rule change moved beyond its margin, no step is attributed to its rule; the 2025 and 2026 labels name the period only. Top-edge intervals read slightly too narrow. Dashes: not estimated for that row.

Table 1. Measured changes in the fitted 50 percent strike contour, 72-inch batter, 2024 pitch mix, read at mid-plate. Cells are point [95% interval]; the placebo row uses its pre-registered 90% interval. The 2023-to-2024 change, with no rule change, exceeded its margin, so the 2025 and 2026 steps are labelled by period and not attributed to a rule.